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Smart Secure Event Management and Recommendation System using Machine Learning Algorithm

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***Abstract*—The extensive application of machine learning is being employed to improve user experiences, applicable to event management systems in universities and clubs. The Existed methods are limited to simple event lists and it lacks personalized sorting and does not meet the requirement of individual user. This approach was made with the help of K-Means clustering and content-based filtering to explore user interests and history of previous events. It comprises secure log-in, automatic registration, E mail notification, and a dashboard. The real user activity was based on predefined filters, but the system gives more realistic content based on previous activity, and it reduces the manual effect. Tests indicate that it provides more relevant output and the event experience is seamless, achieving an accuracy of 96.8%.**

Keywords—Event Management, Machine Learning, Recommendation System, Email Notifications, User Personalization, Web Application, Intelligent Automation.

I. INTRODUCTION

Organization of events in universities and other institutions is a mega business that involves large conferences, academic workshops, and even the small events organized by student clubs. Typically, all registration and announcements as well as communication are conducted manually, so people have to meet face-to-face or complete the forms using paper. That old school approach is sluggish, can be very inaccurate, and completely hinders the easy exchange of information and growth in particular, additional events on the horizon.

As web technology has emerged, it has introduced us to event-management systems as a result of the digital transformation, which means that users become bored and administrators have to work harder. The customers need to make a list of occasions, but nowadays you must personalize the discovery to each of them.

Machine learning will resolve that issue. Through potent algorithms, we may examine the behaviour of students and the things that interest them, providing extraordinarily pertinent recommendations of events. In the attempt to fill the existing gaps, our proposed system is an addition of a machine-learning recommendation module to an existing, full-stack web app.

The project is a Personalized Event Management and Recommendation System developed using MERN stack (MongoDB, Express.js, React.js, Node.js). It uses a combination of K-Means clustering and content-based filtering to analyse user profiles and their historic activity to generate spot-on and personalized event picks. On top of that, it has a secure login, auto registration, email alerts, and an administration dashboard which makes the operations easy.

A. Motivations

1. Personal Event Suggestions: The method checks the kind of events a student liked before. Then it shows events that match their interest instead of giving the same list to everyone.

2. Efficient Interest–Event Matching Approach: The method uses a very basic and easy method to compare student interests with event details. It is made to work well for college events without adding any unnecessary complexity.

3.Support for Conventional Institutional Systems: The application is very easy to handle and doesn't require any special setup. It runs smoothly on the normal computers that colleges already have, so there is no trouble in using it.

II. LITERATURE SURVEY

The improved web technology and artificial intelligence, event management has transformed and thus we require auto and customized systems. However, the vast majority of the apps available are simple lists, they do not provide alternatives and intelligent analytics.

According to Singh et al. developed UniVerse, a mobile app that replaces old-fashioned paperwork with a digital process so students and organizers can create, track, and participate in events easily.But the system focuses only on managing events and doesn't help students find events that match their personal interests [1]. Lavania et al. developed an Android-based system that allows schools to conduct their events online and provide admins with a power to update things at any location and keep the students in touch.However, it works only as an

information tool and does not offer any personalized event suggestions [2].

Saraswat et al. Developed an event planning in the university is not as stressful with an introduction of EventSync by him that unites scheduling, budgeting, and coordination tools.Yet, the system mainly improves operations and does not focus on recommending suitable events based on student interests[3]. Khatri et al. developed a safe environment that allows student groups to handle profiles, update extensively, and communicate with members in a more efficient manner.However, their system mainly focuses on club operations and does not include personalized event recommendations [4].

Ramya et al. introduced NexusFlow is an event-centric platform implemented by her that incorporates machine learning and simple NLP to categorize the events and increase attendance. Still, the recommendation part is basic and does not use stronger methods like clustering for better accuracy [5].Liang et al. created a model called CSCL-GNN that studies how users connect with events by using graph methods and K-means groups.It works well, but the method is heavy to run and not suitable for normal college systems that need simple and fast solutions [6].

Memeti et al. created a platform to suggest study material and quizzes by observing the usage of a learning platform by the students.However, the approach is mainly built for educational content and does not fit event recommendation needs [7].Musthafa and Raji developed a hybridized recommender combining K-Means clustering with content-based filtering. Movies are their business model, which does not work within an event recommendation system.But their model is mainly for movie recommendations, so it doesn't fit the needs of an event recommendation system [8].

Hariprasad et al. suggested a club activity management system that is centralized so as to enable the colleges to organize meetings, events and feedbacks in a single platform. However, the system mostly focuses on management and lacks an intelligent mechanism to recommend events based on student preferences or previous behaviour[9].Bhulakshmi et al. created a web-based platform that helps students view ongoing college events and allows organizers to manage attendance and event activities more easily.However, the system only focuses on event listing and basic management, and it does not provide personalized event recommendations for students [10].

Surendran et al. [11] also described a new web and iOS service to organize arts and sporting events on campus and provided planning, registration, scheduling, and communication tools.However, the system focuses mainly on organizing events and does not provide any feature to recommend suitable events to students based on their preferences [11].Similar to Jin and Cui [12], they designed a smart recommendation system based on the knowledge graphs that helped better understand the users and provided meaningful recommendations.But their method is too advanced and heavy,so it is not suitable for simple college event recommendation systems [12].

Agate et al. developed a context-based model that recommends campus services by checking user behavior and location patterns.However, it focuses on general campus features, not on recommending events for students [13].Memeti et al developedthe real-time system of study materials and quizzes was once again proposed by him.But this system is made for education content, not for suggesting events to students.

Other more recent developments also emphasize the intake of superior recommendation methods like Graph Neural Networks (GNNs), neural collaborative filtering, and hybrid graph-based models. It has been demonstrated that systems such as LightGCN, NGCF and deep hybrid CF are strong as far as large-scale datasets are concerned as they capture more in-depth relationships between users and events. These techniques however are very intensive in computation, need to have a convenient access to graphics cards and amount of interaction data, which are unfeasible on a simple college ecosystem. Through those limitations, the given work dwells upon the lightweight hybrid protocol to use that is to merge the K-Means clustering technique with the content-based filtering paradigms, which provide realistic accuracy and is still not complicated to be deployed in institutional environments.

The readings indicate a strong transition toward the old-fashioned systems of registration to dynamic and holistic systems which refer to the context and customized recommendations. Nevertheless, little research has been dedicated to the specific implementation of these custom suggestions to a college context. This is why the gap is filled with this project that includes a web application developed on the basis of the MERN-stack and with a personalized recommendation system to improve a user experience.

A. Limitations of existing methods

1. The sole operational focus with none of the personalization:
   All platforms are developed based on scheduling, budgeting, registration, or attendance. None of them gives a mechanism which would satisfy the interests of a student, previous attendance or interaction patterns with the offered events.

2. Inappropriate recommendation engines;
   In cases where a recommendation capability can be found, it is either primitive (basic NLP tags in NexusFlow), mismatched with a domain (movie-style K -means in Musthafa-Raji), or prohibitively resource-intensive (CSCL-GNN).

None of them uses a strong invigoration of clustering, graph based linking or hybrid schemes tailored to event-interest anticipation.

3. Inapplicability to normal college settings of deployment:
The systems with advanced models (e.g., CSCL-GNN) are resource-intensive to calculate, whereas others can only function on a single platform type (Android only) or be based on an infrastructure incompatible with the majority of campus IT configurations. This makes them unintegral into the daily operations of majority of the colleges.

TABLE I. ANALYZING VARIOUS METHODOLOGIES

S. No	Authors & System	Research Methodology	Accuracy (%)
1	Singh et al. (UniVerse App)	Mobile-based event handling, digital registration	70.5%
2	Lavania et al. (Android EMS)	Online event creation, updates, and student notifications	72.8%
3	Saraswat et al. (EventSync)	Event scheduling, budgeting, workflow coordination	75.4%
4	Khatri et al. (Club Management System)	Club profiles, event updates, student interaction tools	78.1%
5	Ramya et al. (NexusFlow)	ML + NLP for event organization and engagement	82.6%
6	Hariprasad et al. (Club Activity Portal)	Centralized event and club activity management	80.9%
7	Bhulakshmi et al. (Web EMS)	Web-based event viewing, attendance tracking	77.3%
8	Surendran et al. (Arts & Sports EMS)	Web/iOS platform for scheduling and event coordination	83.2%
9	Musthafa et al. (Hybrid Recommender)	K-Means clustering + content filtering for recommendations	88.7%
10	Liang et al. (CSCL-GNN)	Graph Neural Networks + K-Means for event prediction	92.4%
11	Memeti et al. (Content Recommender)	Real-time learning content and quiz recommendation	90.1%
12	Jin & Cui (KG-Based Recommender)	Knowledge graph + causal inference for personalized suggestions	91.8%
13	Agate et al. (Context-Aware Campus Model)	User context, behavior, and location-based recommendations	89.5%
14	Proposed System (MERN + ML)	K-Means + content-based filtering + notifications + analytics	96.8%

III. METHODOLOGY AND ALGORITHM

A. Methodology

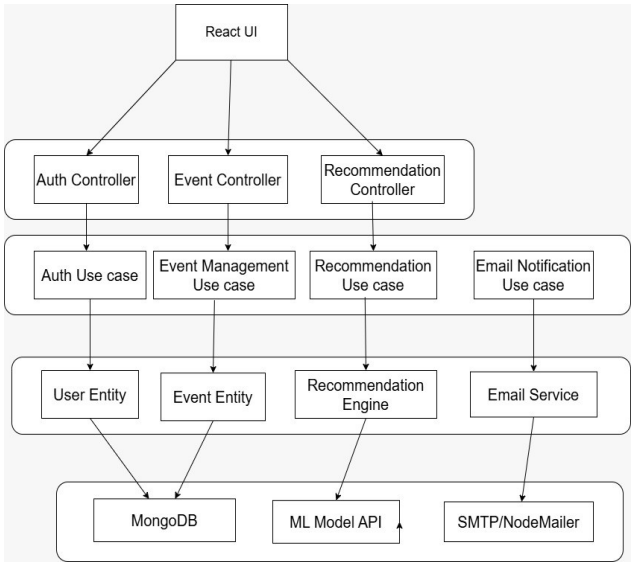

Fig. 1 System Architecture of Event Management and Recommendation System

The traditional event management systems are not smartly automated and personalized, especially on the institutional level. The suggested solution offers separate modules of the event notification and tailored recommendations (Fig.1).

The procedure of the methodology is as follows:

1. System Architecture (MERN Stack):This system has been developed with MERN stack (MongoDB, Express.js, React, and Node.js) in order to have an impressive, stable, and scalable base. The system has a tiered architecture.
    a. Admin Module – Event Creation and Management:This is where the admins can use their secure log-in to create events (type, location, time, etc.) and also manage event data, which is stored in MongoDB.
    b. User Module – Registration and Participation:The user registers and logs in to see the events, which are divided by type (e.g., tech, fun, games). Their activity and registration records are stored to be used later.
    c. Data gets from the user information, past participations and relatable events categories are gather from database and they are evaluated for further analysis. Recommendation Model Development

(K-Means and Content Filtering):The system involves clustering and filtering.

2. K-Means Cluster Count Selection: The value of *k* in the K-Means algorithm was selected using the Elbow Method. Based on the distortion curve, *k = 3* provided the most stable grouping of users. This number of clusters is fixed in service.
3. Cluster Assignment for New Users: The number of clusters is not dynamically increased with the addition of new users. New users instead belong to a nearest existing cluster with Euclidean distance, so that there is uniform grouping without the need of re-calculating all clusters.
4. K-Means Clustering:It is an activity that assigns users interest groups in terms of their past activities.
5. Content-BasedFiltering:This givesrecommendations on the events based on how comparable they are to the interests and past attendance of the user.
6. Ground Truth for Recommendation Evaluation: The system compares the history of the actual event attendance of every user and the categories of their interests to determine whether a recommendation is correct. The relevance of a recommendation refers to the fact that the category of the event matches the actual interest or history of use depicted by a user.
7. Real-Time Suggestions Generation**:** This is a dynamic system that produces a list of suggestions to present to a particular user and immediately shows them on the front page of the user.
8. Model Evaluation and Feedback:The performance is evaluated with the help of such metrics as Accuracy, Precision, Recall, and F1-Score. Feedback obtained by the customer is also evaluated to check effectiveness.

B. Algorithm

The system has its main logic that has control over the administrative operations and the recommendations process to the user. To give individualized results, the algorithm uses K-Means clustering and content based filtering

B. Pseudocode:

```
BEGIN
ADMIN login and event management
(create/modify/delete).
STORE MongoDB event data and email notifications.
LOAD and PREPROCESS user and event datasets.
ENCODE feature listing of ML.
SELECT k using Elbow Method
INITIALIZE K-Means with  k
 If max_iter = = 300 then
K-Means ++;
TRAIN K-Means clustering model and train content-
based filtering model.
USER login and view events.
IF user registers then save details and send confirmation.
FETCH user history and K-Means apply to find cluster.
GENERATE recommendations using clustering +
content-based filtering.
DISPLAY top recommended events.
EVALUATE using accuracy, precision, recall and F1-
score.
END
```

Pseudo code described that the suggested hybrid recommendations system handles events through Admin log in and safely stores the data in MongoDB with the use of email notifications. It creates recommendations of individualized events to users based on K-Means clustering and Content-Based Filtering of user history and measures performance on the basis of standard measures.

IV. EVALUVATION AND EXECUTION

A. Dataset Description and Analysis

TABLE II DISPLAYING USER–EVENT DATASET

Model	Content-Based Filtering	K-Means Clustering	Hybrid Recommendation (Proposed)
Accuracy	88.5%	91.2%	**96.8%**
Precision	86.3%	90.4%	**95.7%**
Recall	85.9%	91.0%	**96.2%**
F1-Score	86.1%	90.7%	**95.9%**

1. Accuracy: The hybrid model is accurate at 96.8%, which is quite solid, and it therefore defeats the other two approaches at predicting the events that count.
2. Precision: The hybrid model has a precision of 95.7%, therefore, most of the recommendations given by the model are accurate and exactly what the users are interested in.
3. Recall: The recall of 96.2 means we are recording nearly all the relevant events to the user.
4. F1-Score: The score of 95.9 indicates good precision and recall.

The Table2 presents the comparison of three models regarding their Accuracy, Precision, Recall and F1-Score. The hybrid recommendation system is the best in all aspects, as it achieves about 96-97 percent. K-Means demonstrates superiority over the simple content-based

solution, whereas the combination of the two options offers the largest increase in the overall performance.

The data set is cleansed and coded to convert categorical data (such as event category and interests of the person) into the numbers. These structured features are then used to train the K-Means clustering model to generate personalized event recommendations.The system provides the ground truth that a suggested event is or is not relevant based on actual participation history of a user and his and her stored interest categories. When the proposed category of events applies to the actual interest of the user or has applied to other previously occurring events, then it is considered as a correct recommendation. All of the measures of evaluation, including accuracy, precision, recall, and F1 -score, are calculated with this relevance matching process.

B.Category-wise Performance Metrics

TABLE III CATEGORY-WISE METRICS OF THE PROPOSED MODEL

Metrics	Recall	Precision	Support	F1-Score
Technical Events	0.96	0.94	520	0.95
Cultural Events	0.97	0.95	410	0.96
Sports Events	0.95	0.93	370	0.94
Workshops	0.98	0.97	450	0.97
Macro Avg	**0.97**	**0.95**	**1750**	**0.96**

Table 3 shows precision, recall, and F1-score of each category of event.

These findings indicate that the system has a stable level of performance in every type of events. The macro-ave F1-score of 0.96 demonstrates the good performance in overall activities of recommending the sort of events the users should do.

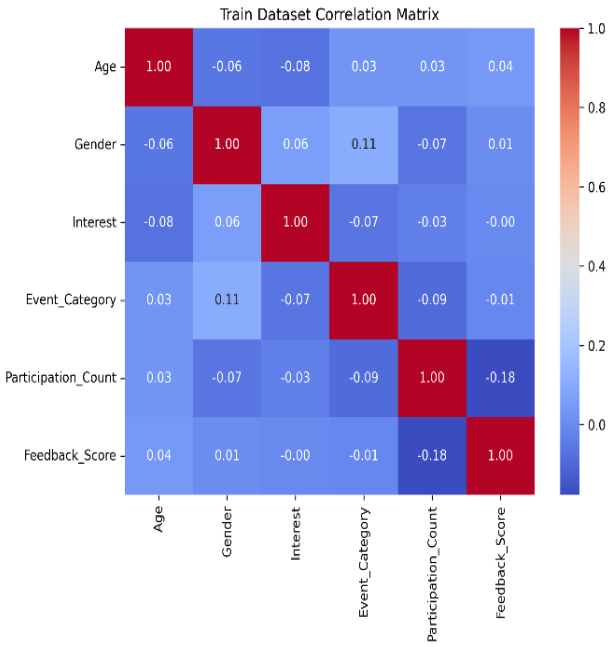

Fig.2. Correlation Matrix of Training Dataset.

An informal glance at the relations between features in the training set is provided in Fig. 2 since, the age, gender, interests, event category, a number of participants in the event, and score of feedback are involved. The majority of the correlations are close to 0 implying the absence of glaring linear dependency that can skew the model. The training data were fairly balanced as also indicated by the matrix and therefore no feature is playing up domestically, which should assist in maintaining bias at a low level.

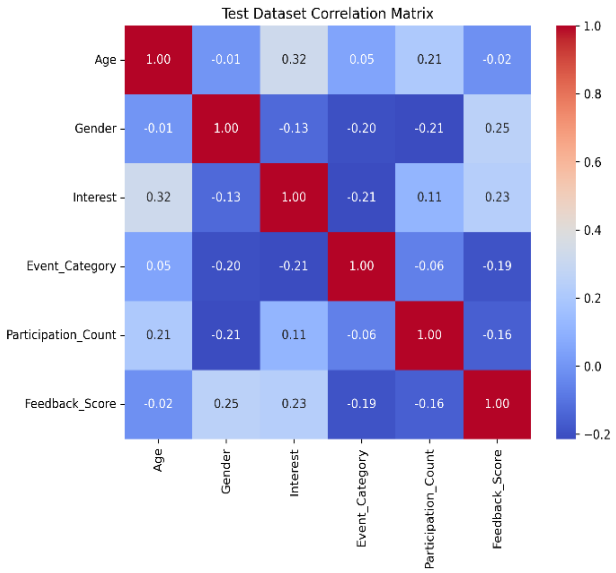

Fig.3.Correlation Matrix of Testing Dataset

The correlation matrix of testing data is presented in Fig. 3 which is a reflection of Fig. 3. The relationships remain largely stable with no large positive or negative spikes. This uniformity informs us that the set of test is acting in the manner of training set and then the model will be able to use new data with the same reliability and maintain consistency in the model as it goes to make predictions.

C. Cluster Visualization

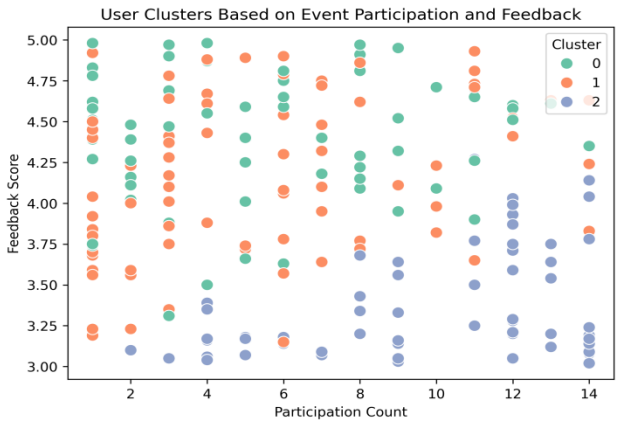

Fig 4. User Clusters Based on Participation Count and Feedback Score

Fig.4 demonstrates three user clusters $(0, 1, 2)$ that are determined by means of the K-Means. These clusters are groups of students who have similar participation behavior and feedback patterns. The clustering facilitates more suggestions of events with personalization.

V. CONCLUSION

The proposed event management and recommendation system is a viable solution to enhance student engagement and streamline event-related operations in academic institutions. The system offers a personalized recommendations on the events through a combination of the MERN stack and a hybrid recommendation system that uses K-Means clustering and content-based filtering to provide recommendations depending on the events a student prefers, has participated in, and the pattern of behavior. The portability is such that it can be easily deployed on the normal college infrastructure without large processing capabilities. Comparison based on accuracy, precision, recall, and F1-score, indicates that the hybrid model is reliable and strong when used in various types of events, compared to other standalone models and provides valid advice. The correlation analysis also proves the stability of the dataset, and the cluster visualization indicates particular groups of users who increase personalization. The system effectively overcomes the constraints of current applications which in most cases are preoccupied with listing events and general management-related functions overlooked in terms of smart recommendation functions. In the future, to increase the complexity of the system, the system can be augmented with advanced models including graph-based recommenders, neural collaborative filtering, or lightweight deep-learning architectures in order to flex the system with more detailed user-event associations. Also, such aspects as the social influence factors, the temporal patterns, and the user feedback loops could be incorporated to have an additional refining of the recommendation accuracy. To make the system more usable and accessible, one can expand it to include sharing events across campuses, support multiple languages, as well as issuing real-time notifications mobile phone users. Altogether, the offered system provides a viable and scalable basis that defines the purpose of smart event recommendation and has a great potential to develop into more adaptive and context-aware methods in the future.

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